

A Novel Image Encryption Algorithm Based On Fractional Fourier Transform and Magic Cube Rotation

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Abstract—This paper proposes a novel image encryption algorithm based on the discrete fractional Fourier transform and an improved magic cube rotation scrambling algorithm. Through fractional Fourier transform and position scrambling, the proposed algorithm can achieve double image encryption in the time-frequency domain. Compared the encrypted images and decrypted images, the proposed image encryption algorithm has better performance than only using fractional Fourier transform.

Keywords—Image Encryption; Fractional Fourier Transform; Image Scrambling; Magic Cube Rotation; Chaotic Sequences

I. INTRODUCTION

Image encryption is the process of using an algorithm to make the transforming information unreadable to anyone except those possessing special knowledge, usually referred to as a key. The fractional Fourier Transform is a relatively new technique. Its complete mathematical definition was first proposed by Namias V. in 1980 [1]. With the development of the research on the discrete fractional Fourier transform, applications on the digital signal processing have also been widely explored.

An image scrambling algorithm based on Rubik's Cube Rotation and Logistic Sequence has been proposed [2]. It assumes if one step indicates a 90 degree rotation, with three middle level of the cube, there are 18 different rotating methods for one step.

A. Chaotic System

In the nonlinear dynamical system, chaos is the certainty of appearance similar to a random process. Mathematically, chaos refers to a very specific kind of unpredictability, which the deterministic behavior is very sensitive to its initial conditions.

One of the most widely used chaotic systems is Logistic system or the Logistic mapping [3], the definition is,

$$x_{n+1} = \mu x_n (1 - x_n) \quad (1)$$

Another chaotic system can be generated by manipulating Chebyshev polynomials,

$$T_n(x) = \cos(n \arccos(x))$$

$$x_{n+1} = T_\mu(x_n) = \cos(\mu \arccos(x_n)) \quad (2)$$

B. Magic Cube Rotation

Rubik's Cube was originally called the "Magic Cube". In a classic Rubik's Cube, each of the six faces is covered by nine stickers, among six solid colors. Every face could be rotated clockwise or counterclockwise.

The original $(3 \times 3 \times 3)$ Rubik's Cube has eight vertexes and twelve edges. In total, there are approximately 4.3×10^{19} permutations. Thus, the large number of permutations leads to a high complexity of the positions on Rubik's Cube.

C. Fractional Fourier Transform(FrFT)

Normally, let $F(f(x))$ denote as the Fourier transform of function $f(x)$, then we have the ath-order fractional Fourier transform of $f(x)$ as $F^a(f(x))$. The definition of the fractional Fourier transforms [4] is

$$F^a(f(x)) = \int_{-\infty}^{+\infty} B_a(x, x') f(x') dx' \quad (3)$$

$$B_a(x, x') = A_\phi \exp[i\pi(x^2 \cot \phi - 2xx' \csc \phi + x'^2 \cot \phi)]$$

$$A_\phi = \frac{\exp\{-i[\pi \operatorname{sgn}(\sin \phi) / 4 - \phi / 2]\}}{\sqrt{|\sin \phi|}}$$

Where a ($0 < |a| < 2$) is fractional order, $\phi = a \cdot \frac{\pi}{2}$.

Besides the previous mentioned definition of the fractional Fourier transform at integral level, there is also a definition at series level [4], which shown as below

$$F^a\{\psi_n(x)\} = \exp(-jna\pi/2)\psi_n(x) \quad (4)$$

$$\psi_n(x) = \frac{2^{1/4}}{\sqrt{2^n n!}} H_n(\sqrt{2\pi}x) \exp(-\pi x^2/2)$$

Where a is fractional order, $\phi = a \cdot \frac{\pi}{2}$.

$$H_n(x) = (-1)^n \times \exp(x^2/2) \frac{d^n}{dx^n} \exp(-x^2/2)$$

$$B_a(x, x') = \sum_{n=0}^{\infty} \exp(-ian\pi/2) \psi_n(x) \psi_n(x') \quad (5)$$

When $a=1$, the fractional Fourier transform is an ordinary Fourier transform.

From the definition, two important characteristics can be deduced [5] as

$$(a) F^{a1} \{F^{a2} \{f(x)\}\} = F^{a1+a2} \{f(x)\}$$

$$(b) F^a \{f(x)\} = F^{a+4n} \{f(x)\}$$

where n is an integer.

D. Discrete Fractional Fourier Transform(DFrFT)

A complete set of eigenfunctions of the fractional Fourier transform is Hermite-Gaussian functions. Candan deduced the discrete Hermite-Gaussian functions, that is DFT matrix Hermite-Gaussian eigenvectors, through the solution of discrete differential equation [6].

Assuming $u_k[n]$ to be an arbitrary orthonormal eigenvector set of the $N \times N$ DFT matrix and λ_k to be the associated eigenvalues, the discrete analog of transform kernel (5) is

$$F^a[m, n] = \sum_{k=0, k \neq (N-1+(N \bmod 2))}^{N-1} u_k[m] \exp(-j\pi k a / 2) u_k[n] \quad (6)$$

$$F^a(f(x)) = \int_{-\infty}^{+\infty} F^a(m, n) f(x) dx \quad (7)$$

II. PROPOSED ALGORITHM

This encryption algorithm uses the discrete fractional Fourier transform to encrypt the image first, then an improved image scrambling algorithm based on magic cube rotation and chaotic sequences has been presented in order to get a better result, compared with only using discrete fractional Fourier transform.

A. Digital Image's DFrFT

The discrete fractional Fourier transform of a digital image is a two-dimensional transform, we could implement it by using the following method:

- Apply discrete fractional Fourier transform to the row vector of the digital image, then get the transform result F1 where the transform order is ax ;
- Apply discrete fractional Fourier transform to the column vector of the digital image, then get the transform result F2 where the transform order is ay ;
- Transpose F2 and the result of two-dimensional discrete fractional Fourier transform can be obtained.

B. Magic Cube Scrambling Algorithm

Assuming one step indicates a 90 degree rotation, there are 30 different rotating methods for one step. Use chaotic systems to generate two pseudo-random sequences to rotate the cubes and get the scrambled result.

- Resize and Partition the original image: Resize the image to the size of $M \times N$ so that the resized image I can be partitioned to 6 parts with no overlap. The sub-images have the size $(M/3) \times (N/2)$. Mark the 6

matrices as Up (U), Down (D), Front (F), Back (B), Left (L) and Right(R);

- Generate the cubes: Pick up a square of size 3×3 pixels from every 6 matrices to attached a cube's six face, and every pixel has a graylevel value. Thus $I_{M \times N}$ could generate $((M \times N) / 54)$ cubes;

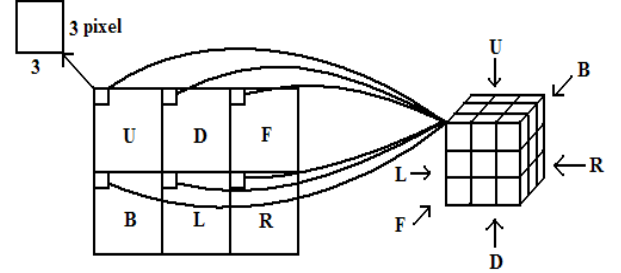


Fig. 1 Mapping the image to a cube

- Construct the rotation method table(RMT_L and RMT_C) Apply (8) (10) with two initial values as secret keys to generate pseudo-random sequences. Either of them converts to a new sequence with integer elements which has the range [1, 30] to indicate rotation methods. Construct RMT_L with L, and RMT_C with C.

$$x_{n+1} = 4x_n(1 - x_n) \quad (8)$$

$$L_i = F(\logistic(Key(1), \frac{M \times N}{54} \times 25)) \quad (9)$$

$$x_{n+1} = \cos(4 \arccos(x_n)) \quad (10)$$

$$C_i = F(\text{chebyshev}(Key(2), \frac{M \times N}{54} \times 25)) \quad (11)$$

- Rotate cubes and remap them to the scrambled result: Rotate every cube according to the corresponding row of RMT_L, and remap the cube back to a intermediate matrix Is. Generate cubes from Is, using RMT_C to rotate cubes and remap them to the scrambled result.

The restoring procedure is similar to the scrambling method. Generate cubes and get rotating method table. Rotate and remap the cubes in a reverse order and reverse direction.

C. A Novel Image Encryption Algorithm

1) Encryption Procedure

Assume the input image as $f(x, y)$, $F^{(ax, ay)}$ is the operator of discrete fractional Fourier transform, where ax is the transform order in x-direction and ay is the transform order in y-direction. $f_a(u, v)$ is the encryption result after discrete fractional Fourier transform. S is the magic cube scrambling function. After the scrambling step, we can get the encrypted result of $Out(u, v)$.

$$\begin{aligned} f_a(u, v) &= F^{(ax, ay)} \{f(x, y)\} \\ out(u, v) &= S(f_a(u, v)) \end{aligned} \quad (12)$$

2) Decryption Procedure

Assume the decrypted image as $f(x, y)$, $F^{(-ax, -ay)}$ is the operator of inverse discrete fractional Fourier transform, where $-ax$ is the inverse transform order in x-direction and $-ay$ is the inverse transform order in y-direction. S^{-1} is the restoring

procedure of image scrambling algorithm based on magic cube rotation and chaotic sequences. The restoring function is as following:

$$\begin{aligned} f_a(u, v) &= S^{-1}\{Out(u, v)\} \\ f(x, y) &= F^{(-ax, -ay)}(f_a(u, v)) \end{aligned} \quad (13)$$

III. EXPERIMENTAL RESULTS AND ANALYSIS

The image encryption algorithm is simulated at the MATLAB 7.4.0 environment. A 256×256 Lena gray image is used as the test examples for image encryption.

A. Results and Analysis of FrFT Encryption

Set $ax = 1.2$, $ay = 1.2$ as the transform orders of discrete Fourier transform to the original image, Fig. 2 (a) .

Fig. 2 (b) shows the encrypted image, which use discrete fractional Fourier transform to encrypt.



Fig. 2 (a) Original Image Fig. 2 (b) Encrypted Image

Fig. 2 (c) shows the decrypted image with the right transform orders for the inverse discrete fractional Fourier transform.



Fig. 2 (c) Decrypted Image Fig. 3 (a) Decrypted Image

Fig. 3 (a) shows the decrypted image with two wrong transform orders where $ax = ay = 0.8$. We can see part of the original Lena image's profile.

Fig. 3 (b) shows the decrypted image with two wrong transform orders for the inverse discrete fractional Fourier transform where $ax = 0.9$, $ay = 0.6$. We can see part of original Lena image's profile.

Fig. 3 (c) shows the decrypted image with two wrong transform orders where $ax = 0.4$, $ay = 1.6$. We can see part of original Lena image's profile.

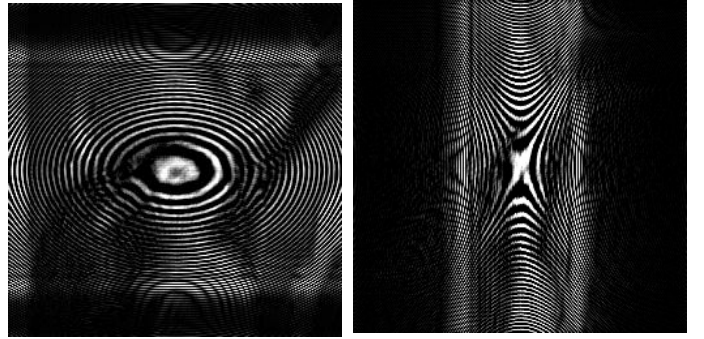


Fig. 3 (b) Decrypted Image Fig. 3 (c) Decrypted Image

As the above figures show, it can be observed that the encrypted image can be successfully decrypted with the right transform orders as secret keys. However, some part of the Lena image can be seen from the decrypted results using incorrect transform orders. Hence, only using fractional Fourier transform to encrypt an image is not safe.

B. Results and Analysis of Double Encryption Algorithm

Set $ax = 1.2$, $ay = 1.2$ as the transform orders of discrete Fourier transform, Key 1= Key 2= 2 as the scrambling keys.

Fig. 4 (b) shows the once encrypted image, which only uses discrete fractional Fourier transform.



Fig. 4 (a) Original Image Fig. 4 (b) FrFT Encrypted Image

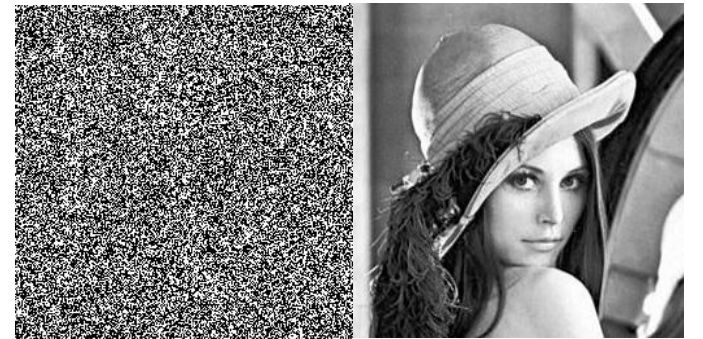


Fig. 4 (c) Encrypted Image Fig. 4 (d) Decrypted Image

Fig. 4 (c) shows the result of the proposed algorithm. Fig. 4 (d) shows the decrypted image with the right keys.

Fig. 5 (a) show the decrypted images with two wrong keys as the transform orders for the inverse discrete fractional Fourier transform.

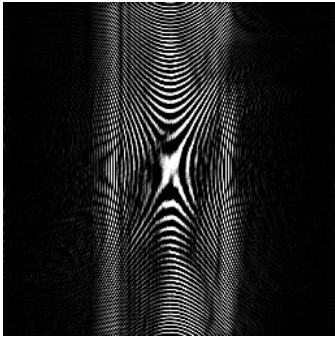


Fig. 5 (a) Decrypted Image
with $ax=0.4$, $ay=1.6$

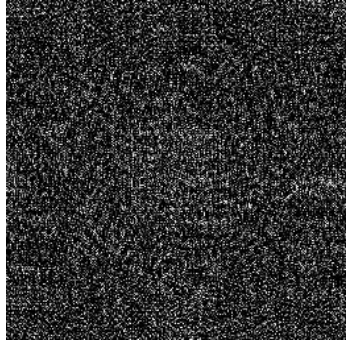


Fig. 5 (b) Decrypted Image
with Key 1=Key 2= 1

Fig. 5 (b) shows the decrypted images with two wrong scrambling restoring keys.

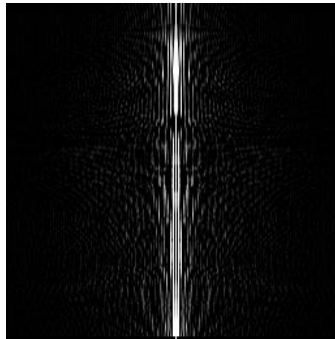


Fig. 5 (c) Decrypted Image
with $ax=0.2$

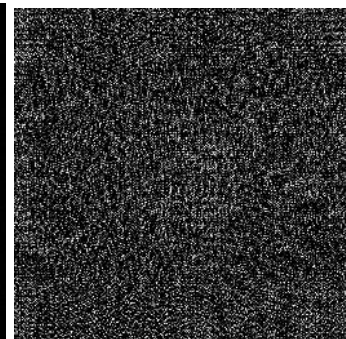


Fig. 5 (d) Decrypted Image
with Key 1 = 1

Fig. 5 (c) shows the decrypted images with a wrong key as the transform order for the inverse discrete fractional Fourier transform. Fig. 5 (d) shows the decrypted images with one wrong scrambling restoring key.

As the above figures show, it can be observed that the encrypted image, using the proposed algorithm, can be successfully decrypted when using four right keys. Compared with Fig. 4 (b), the encrypted result, Fig. 4 (c), has better performance of encryption. For Fig. 5 (b), although the transform orders for the discrete fractional Fourier transform

are correct, if scrambling restoring algorithm's secret keys are wrong, the decrypted result is still an unreadable image.

Therefore, the encrypted image could not be decrypted back to the original image unless with four right keys. Compared with the encryption only using fractional Fourier transform, this algorithm has a large space of keys and a better quality of encryption result.

IV. CONCLUSIONS

In this paper, a novel image encryption algorithm has been presented. Based on discrete fractional Fourier transform and a magic cube scrambling algorithm, this algorithm can implement image encryptions in the time-frequency domain. Through comparing the encrypted image and the decrypted images, the proposed encryption algorithm, which uses discrete fractional Fourier transform encryption and position scrambling, has a better performance than the encryption only using fractional Fourier transform.

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